



Valorizing spent mushroom substrate into syngas by the thermo-chemical process

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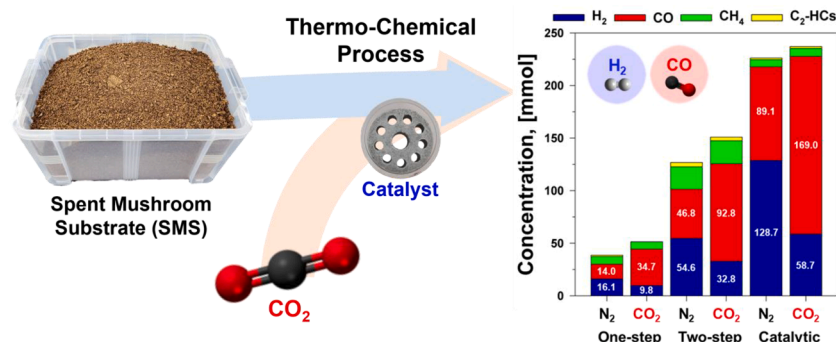
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HIGHLIGHTS

- Spent mushroom substrate was valorized into syngas in the thermo-chemical platform.
- To impart a greener feature to the pyrolysis process, CO₂ was used as reactive gas.
- CO₂ diverts the carbon distribution patterns of pyrogenic products towards syngas.
- Synergistic use of CO₂ and Ni catalyst enhanced the yield of syngas (mainly CO).

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:

Circular economy
Waste-to-energy
Biomass valorization
Pyrolysis
CO₂ utilization

ABSTRACT

This study investigated the conversion of agricultural biomass waste (specifically, spent mushroom substrate) into syngas via pyrolysis. Carbon dioxide was used to provide a green/sustainable feature in the pyrolysis process. All the experimental data highlight the mechanistic role of carbon dioxide (CO₂) in the process, demonstrated by the enhanced carbon monoxide (CO) yield from pyrolysis under CO₂. Carbon dioxide was indeed reactive at ≥ 500 °C. Carbon dioxide was reduced and subsequently oxidized volatiles stemming from the thermolysis of spent mushroom substrate via the gas-phase reaction, thereby resulting in the enhanced formation of CO. Carbon dioxide radically diverted the carbon distribution patterns of the pyrogenic products, as more carbon in the oil was allocated to syngas by the gas-phase reaction of volatiles and CO₂. To enhance the mechanistic role of CO₂, a Ni-based catalyst was added to the pyrolysis process, which greatly accelerated the gas-phase reaction of volatiles and CO₂.

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<https://doi.org/10.1016/j.biortech.2023.130007>

Received 3 October 2023; Received in revised form 27 October 2023; Accepted 9 November 2023

Available online 10 November 2023

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1. Introduction

Society's rate of carbon consumption has accelerated exponentially since the industrial revolution (Ogbonna and Nwoba, 2021). More than 85 % of primary energy is currently obtained from the combustion of fossil fuels (coal, petroleum, and gas) (He et al., 2023; Ozdemir, 2023). Nevertheless, a heavy reliance on fossil fuels is not desirable from the environmental perspective (Wu et al., 2023). Anthropogenic carbon dioxide (CO₂) stemming from fossil fuels disturbs the complex natural cycle of CO₂, and such disturbance is one of the major causes of global warming (Yu et al., 2021). The use of carbon-free/neutral resources (sunlight, wind, tides, and biomass) could be effective in attenuating the severe consequences rising CO₂ levels (Lei et al., 2023). Nonetheless, the use of biomass rather than other renewable resources (e.g., sunlight wind, etc.) in the production of energy is also advantageous considering its energy density (Züttel et al., 2010). Unlike other renewable resources, biomass is carbon-based and offers a great opportunity to seamlessly replace fossil fuels in current distribution networks (Yi et al., 2015). Therefore, biofuel (biodiesel and bioethanol) production has gained considerable attention.

While they are simply blended with fossil fuels (Elkelawy et al., 2022), the conversion process for liquid biofuels faces technical challenges in terms of the degree of carbon utilization. Biodiesel is produced by acid/alkali-catalyzed (trans)esterification of fats (Tien Thanh et al., 2022). Carbohydrates in biomass are only converted into bioethanol during fermentation (Hemansi et al., 2022). This suggests that a limited portion of biomass is converted into liquid biofuel (Ayodele et al., 2020; Carmona-Cabello et al., 2018), while remaining biomass is waste (Yao et al., 2015). Anaerobic digestion (AD) has been widely adopted in an effort to enhance the degree of biomass carbon utilization (Moscoviz and Jimenez, 2021). In fact, three biopolymers (carbohydrates, proteins, and fats) in biomass are transformed into biogas (i.e., methane (CH₄)) through the AD process (Moscoviz and Jimenez, 2021). Nevertheless, the AD process is not free from carbon loss, due to uncontrollable microbial metabolism (cell growth and respiration) (Wang et al., 2022). In addition, a fraction of the aromatics (lignin) in biomass is not biodegradable (Abad et al., 2019). Therefore, it is highly encouraged to seek an applicable technical platform to enhance the degree of carbon utilization in biomass.

Based on these considerations, the thermochemical processes, such as pyrolysis, could offer a breakthrough in the valorization of biomass by enhancing the degree of biomass carbon utilization. Pyrolysis enables the conversion of all carbon in biomass into three-phased pyrogenic products (gas, oil, and char) (Yang et al., 2022). The fraction of carbon in the three pyrogenic products can be controlled by adopting different operational parameters (temperature, reaction medium, residence time, etc.) with or without a catalyst (Mutsengerere et al., 2019). Biocrude (oil) can be used as a fuel and/or as an intermediate in chemical synthesis (Valle et al., 2019). Nevertheless, the complex compositional matrix of biocrude, necessitating complex separation and purification processes, is deemed one of the major obstacles to its widespread use (Tao et al., 2020). Therefore, distributing more carbon into gas could be a strategic approach. Syngas (mixture of hydrogen (H₂) and CO) are the main gaseous products of biomass pyrolysis (Zhu et al., 2022b). The separation and purification of syngas from biomass is easier compared to biocrude (Pena et al., 2020). Furthermore, syngas is already adopted in diverse applications, such as power generation and chemical synthesis (Fiore et al., 2020; Pan et al., 2021; Paykani et al., 2022).

In this context, the main purpose of this study is to seek an

environmentally sound valorization platform for agricultural waste. Spent mushroom substrate (SMS) was chosen as a model for agricultural waste biomass. Mushrooms have gained popularity in response to the rapidly growing demand for a healthy diet (Leong et al., 2022). In 2020 alone, 14.35 million tons of mushrooms were produced (Łysakowska et al., 2023). Approximately 70 million tons of SMS are produced annually (Leong et al., 2022; Williams et al., 2001). The thermochemical process for SMS offers a strategic approach to realizing the concept of waste-to-energy. To impart a green/sustainable valorization platform, the practical use of carbon dioxide as a reactive raw material was investigated in the pyrolysis process. This study laid a particular emphasis on elucidating the mechanistic functionalities of CO₂ in the pyrolysis process. To this end, the thermolytic routes of the SMS were quantitatively and qualitatively characterized. Several test setups for the pyrolysis of SMS were manipulated to investigate the thermolytic route induced by CO₂. Finally, a nickel (Ni)-based catalyst was used for the pyrolysis of SMS to enhance the identified functional roles of CO₂ (from a series of pyrolysis tests). Ni metal was selected due to its greater affordability and catalytic activity for dissociating the chemical bonds in organic materials rather than other ones (Vance et al., 2023). It was hypothesized that the catalytic capability of Ni for activating the C-H and C-C bonds enhances the mechanistic effectiveness of CO₂.

2. Materials and methods

2.1. Chemicals and sample preparation

Spent mushroom substrate was acquired from a commercial mushroom farm (Anseong, South Korea). The SMS was dried at 80 °C for 48 h, then milled and sieved ($\leq 100 \mu\text{m}$). Dichloromethane (DCM; Prod. no. 34856), cellulose (Prod. No. 310697) and lignin (Prod. no. 370959) were purchased from Sigma-Aldrich (St. Louis, MO, USA). Hemicellulose (xylan extracted from corn cobs; Prod. No. X0078) was purchased from Tokyo Chemical Industry (Tokyo, Japan). Two Ni-based catalysts (G90 and G91) were purchased from Clariant International Ltd. (Muttenz, Switzerland). Detailed information on the two catalysts is presented in Table 1. Ultra-high-purity (UHP) grade nitrogen (N₂), carbon dioxide, and air gases were purchased from Green Gas (South Korea). Calibration gas mixture of H₂, CO, CH₄, and C₂-hydrocarbons (HCs) (mixture of ethane (C₂H₆), ethylene (C₂H₄), and acetylene (C₂H₂)) was purchased from INFICON GmbH (Koln, Germany).

2.2. Characterization of SMS

Thermogravimetric analysis (TGA) was carried out to monitor the thermolytic behavior of SMS using a TGA apparatus (STA 449 F5, NETZSCH, Selb, Germany). To thermally degrade SMS, temperature program started from 35 to 900 °C at a constant rate of 10 °C min⁻¹. N₂, CO₂, and air were used to represent the specific atmospheric conditions. The total mass flow rate of the flow gases was set as 100 mL min⁻¹ (consisting of 80 mL min⁻¹ of atmospheric gas (N₂, CO₂, or air) and 20 mL min⁻¹ of balance protective gas (N₂)). All the TGA tests were performed with a SMS loading of 10.00 $\pm$ 0.01 mg. Proximate analysis of SMS (dry basis) was performed according to the standard protocols (ASTM D7582-15, ASTM D3175-11, and ASTM D3174-12). Mass of moisture, volatile matter, fixed carbon, and ash were 5.8, 68.8, 20.1, and 5.3 wt%, respectively. Ultimate analysis of SMS was conducted using the elemental analyzer (EA3000, EuroVector, Italy). Mass of carbon, hydrogen, oxygen, and nitrogen were 44.2, 6.0, 36.3, and 0.8 wt%,

Table 1
Detailed information on the two Ni-based catalysts.

No.	Catalyst	Trade Name	Support	NiO Content,[wt.%]	K ₂ O Content,[wt.%]	Shape	Size,[mm]
1	G90	ReforMax® 330	CaAl ₁₂ O ₁₉	14		10-hole rings	19 x 16
2	G91	ReforMax® 210	CaK ₂ Al ₂₂ O ₃₄	18	1.5	10-hole rings	19 x 12

Table 2
Detailed testing parameter information for analysis of the micro-GC.

	Channel A	Channel B
Column information	Molsieve 5A (10 m × 320 μm × 30 μm)	Plot U (8 m × 320 μm × 30 μm)
Injection time, [msec]	30	30
Backflush time, [sec]	10	5
Injector temperature, [°C]	80	80
Column temperature, [°C]	100	80
Column pressure, [psi]	25	17

respectively. To determine the inorganic mass content of the SMS, inductively coupled plasma-optical emission spectroscopy (ICP-OES, 5800, Agilent Technologies, Inc., Santa Clara, CA, USA) was used.

2.3. Pyrolysis test setups

Pyrolysis tests of the SMS were performed using three different systems (one-step, two-step, and catalytic pyrolysis) developed in the previous study (Lee et al., 2020). For one-step pyrolysis test, a quartz pipe (250 mm outer diameter, 3 mm thickness, and 600 mm length) was used. The tubular reactor was heated using a tubular furnace, which was programmed from 100 to 700 °C at a constant rate of 10 °C min⁻¹. For two-step pyrolysis test, a tubular furnace was additionally located adjacent to the first, which was isothermally controlled at 700 °C. Catalytic pyrolysis was conducted using two Ni catalysts (G90 and G91). Two pieces of each catalyst were alternately loaded in the middle of the second furnace. For all pyrolysis tests, the SMS (10.00 ± 0.01 g) was loaded in the center of the first furnace. The total gas flow rate for N₂ and CO₂ was 800 mL min⁻¹. The concentration of CO₂ for the tests was set to 80 vol% (balanced with N₂) for comparability to the experimental findings from the TGA.

2.4. Analyses of the liquid/gaseous pyrogenic products

During pyrolysis, volatiles were generated by SMS thermolysis. To condense them, the pyrogenic products were cooled at -40 °C using a chiller. All chemical compounds in the condensate were identified using gas chromatography/mass spectrometry (GC/MS, 8890/5977B, Agilent). The non-condensable (gaseous) pyrogenic products were directly linked to a micro-gas chromatograph (3000A, INFICON). Concentrations of the gaseous products were measured every 3 min with the

injection volume of 5 μL. The micro-GC was constructed with two columns (Molsieve 5A and Plot-U). Detailed testing parameter information for analysis of the micro-GC is summarized in Table 2. The main pyrogenic gases (H₂, CO, CH₄, and C₂-HCs) that evolved from a series of SMS pyrolysis tests were measured by micro-GC.

3. Results and discussion

3.1. Characterizing the thermolytic behaviors of SMS

Substrates provide essential nutrients for fungal growth in commercial mushroom cultivation (Carrasco and Preston, 2020). Mushroom substrates are produced using various agricultural residues (sawdust, rice husks, rice straw, wheat/rice bran, etc.) (Hultberg et al., 2023). As such, the residues used could contain large amounts of cellulose, hemicellulose, and lignin (Li et al., 2021). Approximately 20 wt% of the substrate is assimilated by the mushrooms, while the rest remains as waste (SMS) (Meng et al., 2019). Nevertheless, SMS still contains large amounts of cellulose, hemicellulose, and lignin. TGA tests of these individual compounds were performed under inert gas conditions, and their mass changes and thermolytic rates from 35 to 900 °C are given in Fig. 1(a), and TGA of SMS was performed under identical operational conditions. The mass change of the SMS and its thermolytic rate are shown in Fig. 1(b).

The thermal stabilities of cellulose, hemicellulose, and lignin were determined by the temperature showing the fastest thermolytic rate, as shown in Fig. 1(a). The temperature window, which indicates the occurrence of a distinctive mass change, could also indicate the thermal stability of these compounds. As shown, the mass change of cellulose under inert gas conditions was observed from 250 to 400 °C, and its thermolytic rate reached the highest level at 340 °C. Moreover, the temperature window delineating the distinctive mass change of cellulose with respect to lignin was much narrower. Given that cellulose comprises a repeating unit of glucose (C₆H₁₂O₆), such a narrow temperature window is attributed to thermally induced rapid volatilization by bond scissions of a monomer from the polymeric backbone (Olszewski et al., 2019). The characteristic mass change of hemicellulose was achieved from 180 to 340 °C. This suggested that the thermal stability of hemicellulose was inferior to that of cellulose. Interestingly, the thermolytic rate of hemicellulose showed distinctive peaks at 240 and 290 °C, indicating that the thermolysis of hemicellulose followed two distinctive degradation routes. The degradation of hemicellulose was more complex than that of cellulose, and it was difficult to elucidate the complex thermolytic pattern at this stage of the work since the TGA experiment only revealed the mass alternation of hemicellulose when increasing the thermolytic temperature.

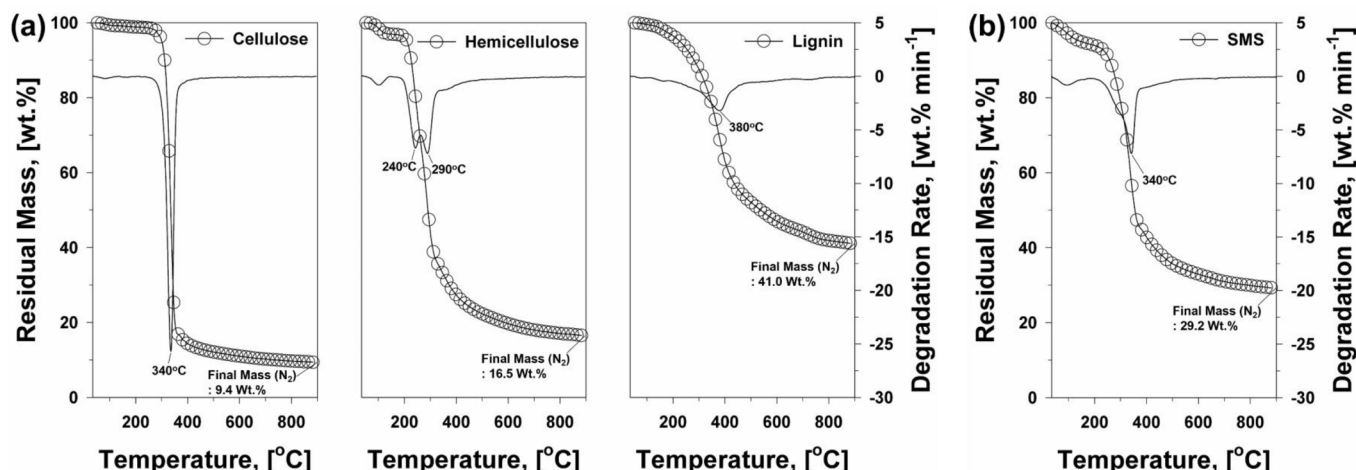


Fig. 1. Mass change and its degradation rate of (a) cellulose/hemicellulose/lignin and (b) spent mushroom substrate (SMS) under inert gas condition.

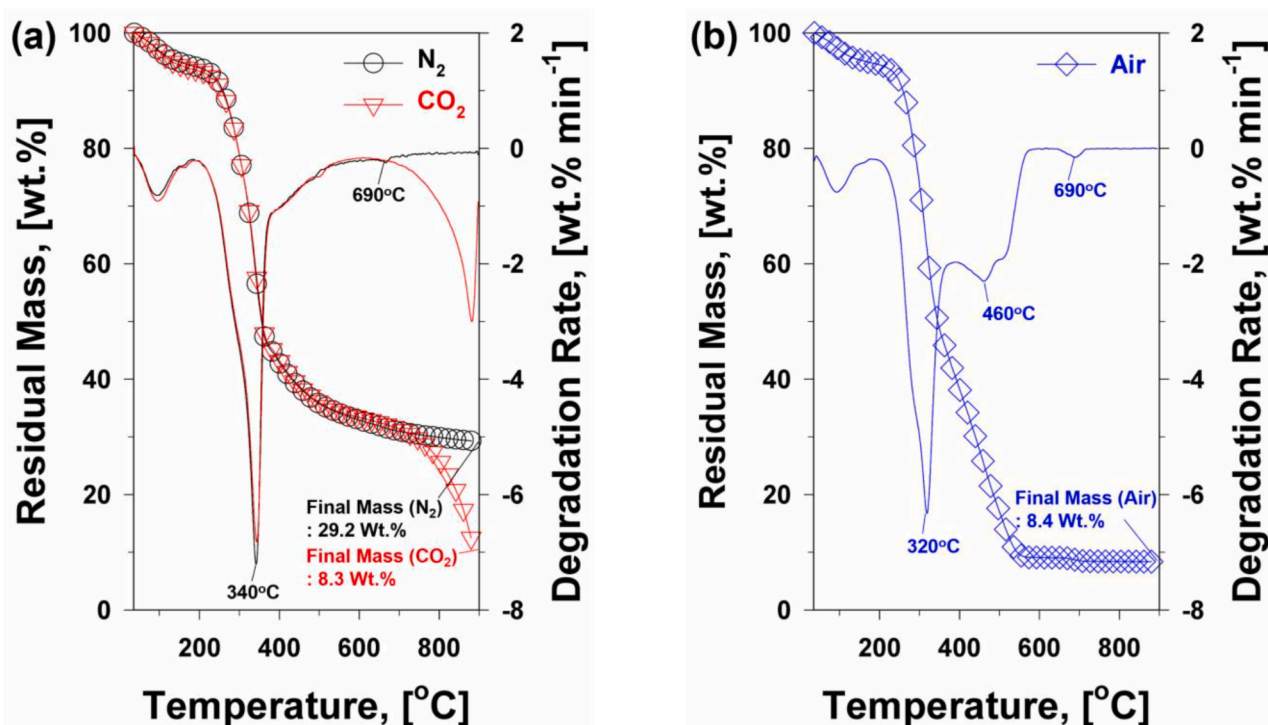


Fig. 2. Mass change of SMS and its degradation rate under (a) N_2/CO_2 , and (b) air.

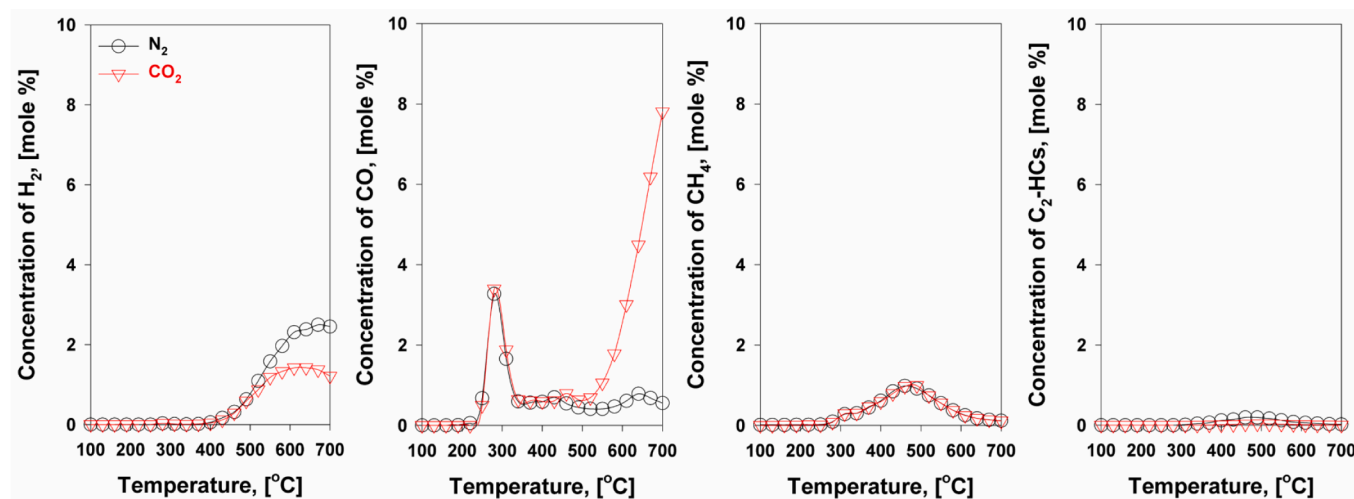


Fig. 3. Concentration profiles of the main pyrogenic gases (H_2 , CO , CH_4 , and C_2 -HCs) recovered from SMS under N_2/CO_2 .

The thermolysis of lignin occurred from 180 to 450 °C, and the fastest thermolytic rate was developed at 380 °C. Therefore, it was concluded that the thermal stability of lignin is superior to those of cellulose/hemicellulose. Considering that lignin comprises phenolic analogs (Singh et al., 2021), there is a rational basis for the superior thermal stability of lignin compared to those of cellulose/hemicellulose. It has in fact been suggested that the aromaticity of lignin imparts thermal stability (Zhu et al., 2022a). The thermolytic rate of lignin at ≥ 420 °C was retarded, which is likely due to dehydrogenation. Indeed, the dehydrogenation kinetics are proportional to the temperature (Lai and Barnard, 2012). In addition, it has been suggested that this retarded thermolytic rate could be key for biochar formation (Zheng et al., 2022). Similarly, biochar formation was observed during the thermolysis of (hemi)cellulose. Nevertheless, the formation of biochar from the thermolysis of lignin was greater than that from (hemi)cellulose.

The thermolytic behavior of SMS under inert gas conditions is shown in Fig. 1(b). The mass change of SMS (64.2 wt%) was completed at ≤ 500 °C. This is comparable to the mass of volatile matter (68.8 wt%) obtained from the proximate analysis of SMS. Interestingly, the thermolytic rate of SMS was the fastest at 340 °C, identical to that of cellulose. Therefore, it is suggested that the volatiles from SMS were mostly released from the thermolysis of cellulose. In addition, the temperature at which the thermolysis of SMS was started was lower than that of cellulose. This can be attributed to the thermolysis of hemicellulose. Nevertheless, the lack of discernible thermolytic rate peaks at 240 and 290 °C indicated that SMS does not contain a large amount of hemicellulose. In addition, the residual mass of the SMS at 900 °C (29.2 wt%) was higher than that of (hemi)cellulose. This showed a good agreement with the result (25.4 wt%) from the summation of fixed carbon and ash contents in the proximate analysis of SMS. Thus, it was concluded that

Table 3

Peak areas of the main constituents in oils from pyrolyses of SMS under N₂/CO₂ (# difference = [(area in CO₂) – (area in N₂)] × (area in N₂)⁻¹ × 100).

No.	Retention Time, [min]	Chemical Constituents	Peak Area in N ₂ , [Abs. × 10 ⁶]	Peak Area in CO ₂ , [Abs. × 10 ⁶]	Difference [#] , [area%]
1. Nitrogen Compounds					
1	8.64	pyridine	4.92	3.85	–21.77
2	12.05	3-methylpyridine	5.38	3.53	–34.36
3	36.39	pyridin-3-ol	6.96	7.10	2.03
		Sub total	17.26	14.48	–16.10
2. Furanic Compounds					
4	13.68	cyclopent-2-en-1-one	5.04	3.34	–33.83
5	14.24	1-hydroxybutan-2-one	4.11	2.66	–35.26
6	16.41	acetic acid	25.22	14.55	–42.28
7	16.67	furan-2-carbaldehyde	11.99	8.08	–32.64
8	17.77	1-(furan-2-yl)ethan-1-one	3.25	2.26	–30.33
9	17.97	allyl butyrate	3.69	2.47	–33.04
10	21.66	furan-2-ylmethanol	21.25	14.79	–30.42
11	23.57	furan-2(5H)-one	4.75	3.12	–34.27
12	24.08	cyclopentane-1,2-dione	7.60	3.64	–52.11
13	25.39	3-methylcyclopentane-1,2-dione	16.87	10.81	–35.93
14	26.81	3-ethyl-2-hydroxycyclopent-2-en-1-one	5.30	3.26	–38.45
15	37.37	5-(hydroxymethyl)dihydrofuran-2(3H)-one	4.12	2.57	–37.77
		Sub total	113.19	71.55	–36.79
3. Phenolic Compounds					
16	26.00	2-methoxyphenol	30.20	21.78	–27.88
17	27.99	2-methoxy-4-methylphenol	11.60	7.77	–33.05
18	28.95	phenol	20.07	14.28	–28.83
19	29.48	4-ethyl-2-methoxyphenol	11.32	7.88	–30.39
20	30.46	4-methylphenol	22.78	15.56	–31.69
21	30.60	3-methylphenol	7.55	5.22	–30.86
22	32.06	4-allyl-2-methoxyphenol	3.66	3.17	–13.21
23	32.21	3-ethylphenol	5.17	4.02	–22.19
24	32.56	2-methoxy-4-vinylphenol	16.08	10.51	–34.62
25	35.25	2-methoxy-4-(prop-1-en-1-yl)phenol	14.95	10.06	–32.70
26	35.33	2,6-dimethoxy-4-methylphenol	26.24	17.42	–33.64
27	36.09	1,4:3,6-dianhydro-α-D-glucopyranose	6.68	4.39	–34.20
28	36.32	4-ethyl-2,6-dimethoxyphenol	25.14	17.71	–29.56
29	37.91	3-methoxybenzene-1,2-diol	9.22	5.34	–42.03
30	38.49	4-allyl-2,6-dimethoxyphenol	6.82	4.68	–31.34
31	38.99	2,6-dimethoxy-4-vinylphenol	28.77	19.28	–32.99
32	39.76	(E)-2,6-dimethoxy-4-(prop-1-en-1-yl)phenol	5.69	4.16	–26.93
33	39.97	1-(3-hydroxy-4-methoxyphenyl)ethan-1-one	3.72	3.10	–16.76
34	40.24	4-butyl-2-methoxyphenol	8.39	6.07	–27.65
35	40.94	pyrocatechol	18.04	12.13	–32.73
36	41.43	4-allyl-2,6-dimethoxyphenol	45.18	31.92	–29.35
37	45.32	4-(3-hydroxypropyl)-2-methoxyphenol	4.50	3.37	–25.12

Table 3 (continued)

No.	Retention Time, [min]	Chemical Constituents	Peak Area in N ₂ , [Abs. × 10 ⁶]	Peak Area in CO ₂ , [Abs. × 10 ⁶]	Difference [#] , [area%]
38	45.73	1-(4-hydroxy-3,5-dimethoxyphenyl)ethan-1-one	8.63	6.15	–28.73
39	46.00	4-(4-hydroxy-3,5-dimethoxyphenyl)butan-2-one	18.54	13.15	–29.04
40	46.16	hydroquinone	5.96	3.92	–34.20
		Sub total	364.87	253.05	–30.65
		Total	495.32	339.08	–31.54

SMS contains a large amount of lignin.

To determine the functional role of CO₂ in the pyrolysis test, TGA of SMS was performed under CO₂ environments. In Fig. 2(a), the mass change of the SMS and its thermolytic rate are plotted as increasing the test temperature. For comparison, the TGA results obtained under CO₂ conditions were overlaid with those under the N₂ condition.

The trends in the mass reductions of SMS under both gas environments were identical at ≤ 690 °C (Fig. 2(a)). However, the trends in the mass reduction of SMS under the CO₂ condition at ≥ 690 °C were different from those under the inert gas environment (N₂). More mass reduction from the CO₂ condition was developed at ≥ 670 °C. Accordingly, the residual mass of SMS at 900 °C under the CO₂ condition was 8.3 wt%. In comparison, the residual mass of SMS at 900 °C from the N₂ environment was 29.2 wt%. The differences in the final masses of the SMS C under both gas environments are likely due to the Boudouard reaction, which enables the conversion of carbon in biochar into CO, which spontaneously occurs at ≥ 710 °C (Lahijani et al., 2015). It was presumed that the appearance of the Boudouard reaction at ≥ 690 °C was likely due to the catalytic effectiveness of earth alkaline metals in the SMS (Jing et al., 2013). To confirm this, the inorganic content of the SMS was determined using ICP analysis. The SMS was found to contain 0.36 wt% Ca, 0.24 wt% Mg, and 0.13 wt% K. Thus, it was theorized that earlier appearance of the Boudouard reaction was catalytically expedited by the earth alkaline metals in SMS. To confirm the degree of redox by the Boudouard reaction, TGA test of the SMS was carried out in air. The final masses of the SMS from CO₂ and air were nearly identical, suggesting that most of the carbon in the biochar was converted to CO by means of the Boudouard reaction.

At ≤ 690 °C, the thermolysis trends for SMS from N₂/CO₂ as shown by TGA are similar. Therefore, it could be concluded that CO₂ did not participate in the process. In other words, the equivalent trends in the thermolytic mass changes of SMS from both two gas environments in Fig. 2(a) inferred an inertness of CO₂ at ≤ 690 °C. Nevertheless, such data interpretation is only pertinent to heterogeneous reactions of CO₂ with the SMS. In short, the functionality of CO₂ on the homogeneous reaction (i.e., gas-phase reaction (GPR)) with volatiles stemming from the thermolysis of SMS could not be fully evaluated because TGA only provides information on the mass alteration of SMS at the corresponding test temperature (Felix et al., 2022).

3.2. Pyrolysis of SMS under the CO₂ condition

To determine the functional role of CO₂ using GPR, pyrolysis tests of SMS were performed under two gas conditions. In detail, SMS was pyrolyzed in the temperature window of 25 to 700 °C at a constant rate of 10 °C min⁻¹. The main pyrogenic gases (H₂, CO, CH₄, and C₂-HCs) evolved from SMS were monitored as increasing the test temperature. The concentration profiles are plotted in Fig. 3.

The main pyrogenic gases recovered from SMS under inert gas conditions follow general evolutionary trends in the biomass pyrolysis. In short, the formation of H₂ was began at ≥ 420 °C by means of

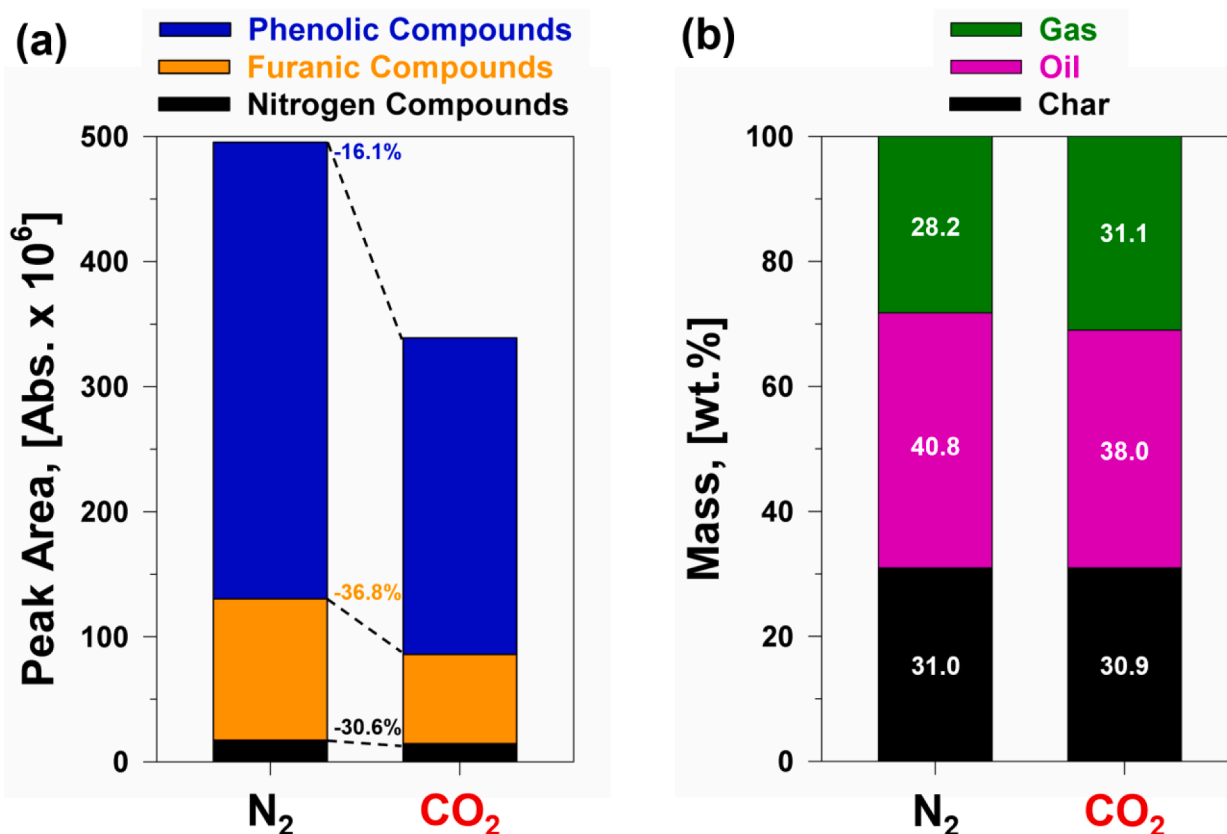


Fig. 4. (a) Total peak area of the three chemical groups (phenolic/furanic/nitrogen compounds) in oils and (b) overall mass balances of the pyrogenic products (gas, oil, and char) from pyrolyses of SMS under N_2/CO_2 .

dehydrogenation. Given that the degree of dehydrogenation is in proportion to the pyrolytic temperature (Lai and Barnard, 2012), this reaction is consistent with the concentration profile of H_2 (Fig. 3). In addition, the H_2 formation of was in good accordance with the TGA results for SMS (for example, the degradation rate of SMS was delayed at $\geq 420^\circ C$). The main mechanism governing biochar formation is dehydrogenation (Lai and Barnard, 2012; Zheng et al., 2022). The CO formation began to increase at $220^\circ C$, and the highest CO concentration (3.8 mol%) was achieved at $280^\circ C$. The CO formation was likely due to the deoxygenation of the (hemi)cellulose polymeric backbone (Olszewski et al., 2019). Despite the low concentration levels of CH_4 and C_2H_6 s, their formation could also be attributed to the thermal cracking of cellulose, hemicellulose, and lignin in the SMS (Amutio et al., 2012).

Interestingly, the concentration profiles of the main pyrogenic gases at $\geq 500^\circ C$ from both gas conditions were indeed different. Specifically, a noticeable increase in CO from the pyrolysis of SMS under CO_2 conditions was observed. This enhanced formation of CO at $\geq 500^\circ C$ could not be governed by the Boudouard reaction because its occurrence was achieved only at $\geq 690^\circ C$, based on the results of the TGA (Fig. 2(a)). Given that the formation of H_2 was suppressed by the CO_2 condition, the reverse water gas shift (rWGS) reaction is a potential mechanistic route to explain this enhanced CO production (González-Castaño et al., 2021). Nevertheless, the enhanced CO formation could not be fully explained by the rWGS, because it produces 1 mol of CO by stoichiometrically consuming 1 mol of H_2 (González-Castaño et al., 2021), while the observed enhanced CO formation far exceeded the suppressed H_2 formation. It was therefore recognized that the enhanced CO production could be achieved without an additional source of carbon and oxygen. As such, the enhancement of CO could be solely attributed to CO_2 through the GPR with volatiles stemming from SMS thermolysis. In other words, it is hypothesized that CO_2 acts a vital role as a carbon and oxygen donor, enhancing the formation of CO via GPR.

As such, GPR can provide a favorable opportunity in modifying the compositional matrix of pyrogenic products. To confirm this hypothesis, the specific chemical components in the pyrolysis oils from the SMS pyrolysis under the two gas conditions were characterized qualitatively. GC/MS analyses of the oils enabled the identification of the chemical species (GC/MS chromatograms not shown; identifications with the peak areas of main chemical species are given in Table 3). The differences in the peak areas of the main species of the oil from two gas environments likely delineate the compositional modifications induced by the GPR. The main constituents of the oils were sorted based on their analogs (phenolic/furanic/nitrogen compounds). The total peak area of each chemical group in the oils (from the SMS pyrolysis under N_2/CO_2) is shown in Fig. 4(a). The total mass balances of the three-phased pyrogenic products (gas, oil, and biochar) from N_2/CO_2 are presented in Fig. 4(b). As summarized in Table 3, the main chemical compounds involved in pyrolysis oil from the inert gas condition are mainly phenolic/furanic analogs. The total peak area of the phenolic compounds was three times larger than that of the furanic compounds. This result is in good agreement with the TGA results (Fig. 1(a)). In Section 3.1, it was noted that SMS is mainly composed of lignin with a small mass fraction of (hemi)cellulose. The thermolysis of (hemi)cellulose generates furanic analogs (Ansari et al., 2019), whereas phenolic compounds are primarily derived from lignin. The peak area of nitrogen compounds in oil was negligible, which is associated with the trace amount of nitrogen (0.8 wt%) measured from the ultimate analysis of SMS. Interestingly, the peak areas of the major chemical species in pyrolysis oil resulted from the CO_2 condition were reduced compared with those under the N_2 condition. Nevertheless, this phenomenon is case-sensitive. The concentration of furanic analogs under the CO_2 condition was reduced by 36.8 % compared to that obtained under inert gas conditions. Although phenolic analogs were the main fraction in oil, their reduction (16.1 %) by CO_2 was lower than that of furanic analogs.

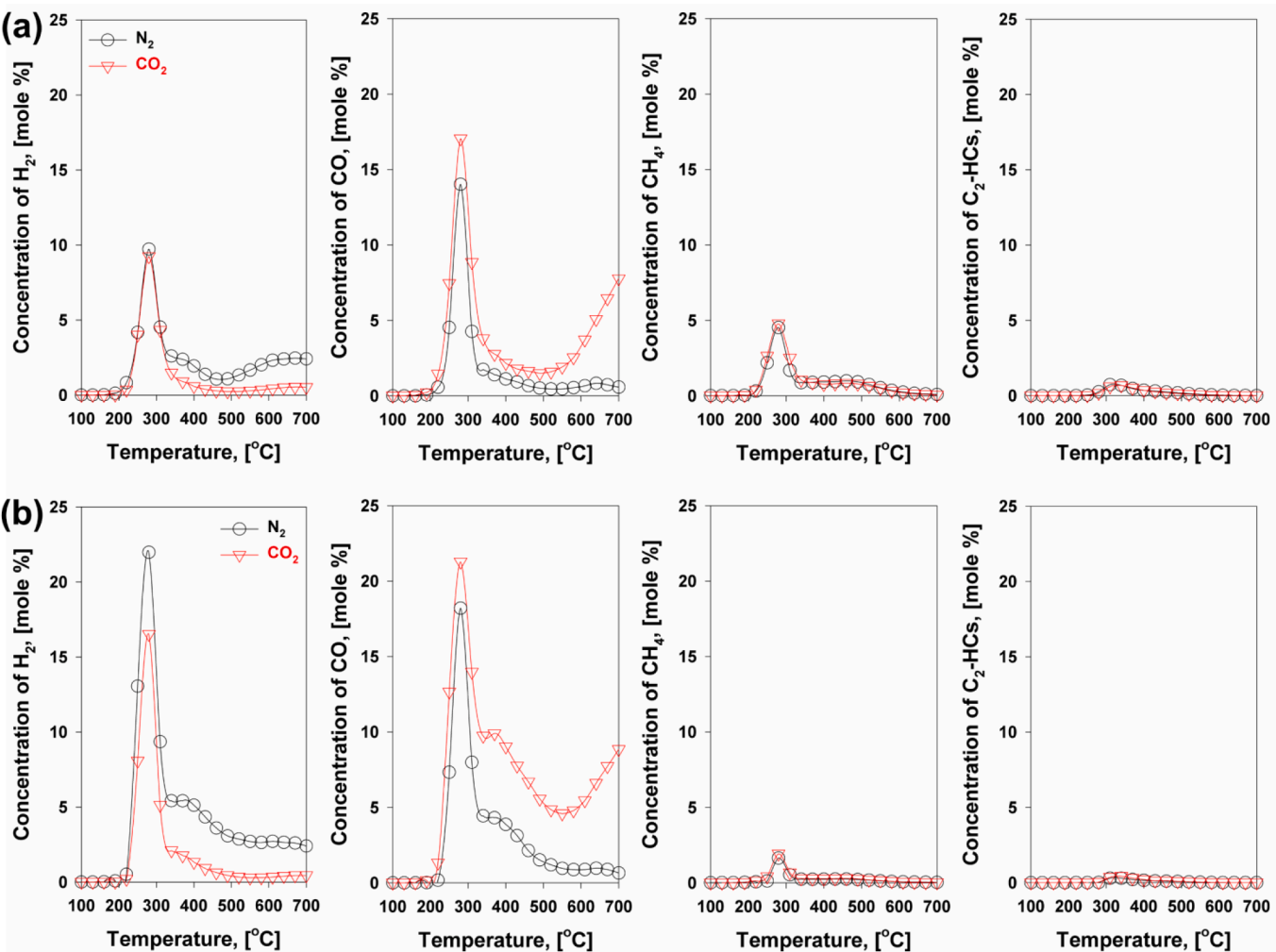


Fig. 5. Concentration profiles of the main pyrogenic gases (H_2 , CO , CH_4 , and C_2 -HCs) from (a) two-step pyrolyses of SMS and (b) catalytic pyrolyses of SMS under N_2/CO_2 .

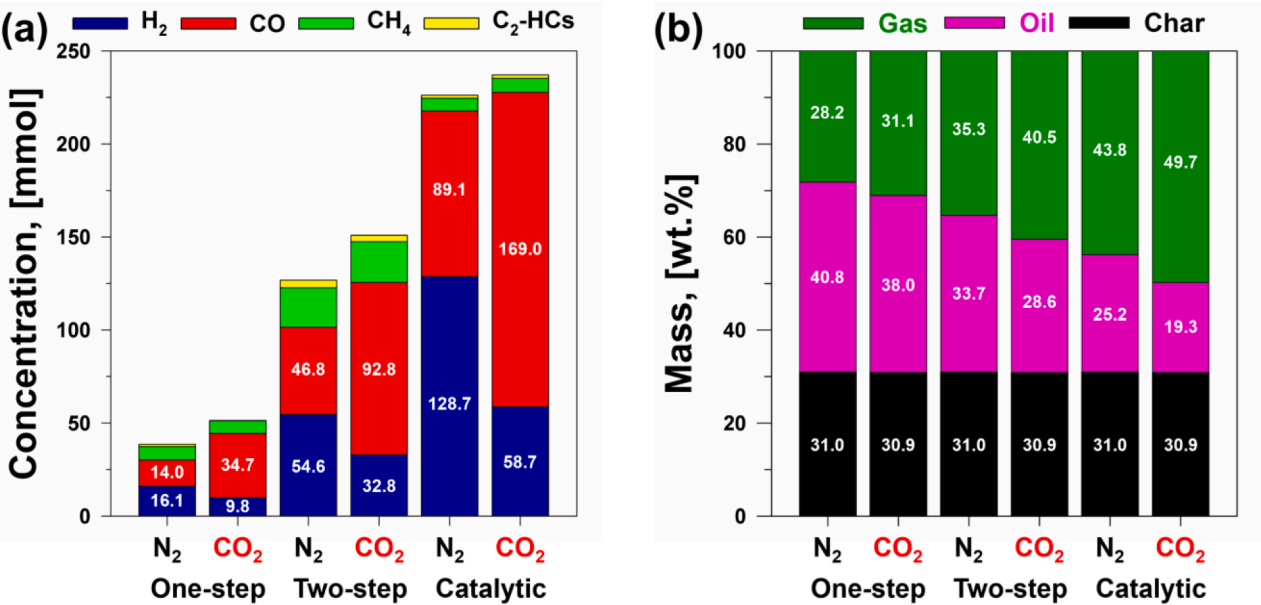


Fig. 6. (a) Cumulative concentrations of the main pyrogenic gases (H_2 , CO , CH_4 , and C_2 -HCs) and (b) overall mass balance of the three pyrogenic products (gas, oil, and char) obtained from one-step/two-step/catalytic pyrolyses SMS under N_2/CO_2 .

This could explain why the resonance structure of the phenolic analogs makes thermal cracking difficult (Hernandez-Castillo et al., 2022). Considering the reduced formation of phenolic and furanic compounds, it was assumed that CO₂ led to the partial oxidation. Thus, the GPR induced by CO₂ enables the conversion of volatiles into CO. This explanation is supported by the mass balance results drawn in Fig. 4(b).

3.3. Two-step/catalytic pyrolysis of SMS under the CO₂ condition

It was experimentally confirmed that CO₂ led to the increased production of CO by GPR. Indeed, the enhanced CO production was developed at ≥ 500 °C, although volatiles from SMS were largely evolved at ≤ 500 °C. To enhance the CO formation by GPR of volatiles, the pyrolysis test setups were modified by adding an additional furnace (i.e., two-step pyrolysis). In detail, volatiles stemming from SMS were isothermally heated at 700 °C. The concentration profiles of syngas and C₁₋₂-HCs from the two-step pyrolyses of SMS under two gas atmospheres are plotted in Fig. 5(a).

Under inert gas conditions, the general trends in the evolution of the main pyrogenic gases between the one- and two-step pyrolysis of SMS were very similar. Nonetheless, enhanced formations of syngas, CH₄, and C₂-HCs were observed between 200 and 350 °C. This is mainly derived from the enhanced thermal cracking of volatiles. Under CO₂ conditions, the enhancement of CO was observed at all test temperatures. This suggests that the reaction kinetics of the GPR of CO₂ with volatiles was promoted. To enhance the reaction kinetics of GPR induced by CO₂, two Ni-based catalysts were used. G90 and G91 are well-known reforming catalysts that enhance syngas production (Fernandez et al., 2022). The concentration profiles of syngas, CH₄, and C₂-HCs are plotted in Fig. 5(b).

The presence of Ni-based catalysts during the pyrolysis of SMS under N₂ enhanced the formation of syngas. Compared to two-step pyrolysis, the concentrations of H₂ and CO increased by 2.3 and 1.3 times, respectively. The enhanced H₂ formation was due to the catalytic dehydrogenation of volatiles. Unlike two-step pyrolysis of SMS, catalytic pyrolysis under CO₂ conditions concurrently enhanced the CO formation and suppressed the H₂ formation. This was likely due to the occurrence of rWGS. Nevertheless, the enhanced yield of CO far exceeded the suppressed formation of H₂ at all test temperatures. Thus, GPR induced by CO₂ resulted in the catalytically enhanced formation of CO.

To compare the effectiveness of CO₂ in all the experimental setups, the cumulative concentrations of the main pyrogenic gases are presented in Fig. 6(a). The functional role of CO₂ in the one-step, two-step, and catalytic pyrolysis led to an increased yield of syngas (mainly CO). As such, the use of CO₂ provides an opportunity to utilize more carbon in thermochemical processes. The enhanced production of CO induced by CO₂ effectively altered the molar ratio of H₂ to CO. A controllable H₂/CO ratio allows for the practical and commercial applications. Syngas can be used in the synthesis of alcohols and a variety HCs, equivalent to the H₂/CO ratio of ~ 1 and ~ 2 , respectively (Khodakov et al., 2007). Using CO₂ in the catalytic pyrolysis of SMS manipulates the H₂/CO ratio in the range of 1.44 to 0.35. When the WGS reaction occurs in the following unit operation, a noticeable increase in the H₂/CO ratio can be realized. The entire conversion of CO by the WGS offers a chance for the improved selectivity of H₂ appearing ≥ 96 vol%. The high purity H₂ could be applied for the fuel cell and/or power generation. The catalytic pyrolysis of SMS and the subsequent conversion of CO into H₂ has a potential to produce 22.8 mmol H₂ g_{SMS}⁻¹ (Fig. 6(a)). Considering that annual generation of SMS is 70 million tons, the estimated production of H₂ stemming from SMS could be 3.2×10^6 kg H₂. Thus, it is estimated that the H₂ production from the catalytic pyrolysis of SMS offers an economic benefit corresponding to 32 million USD, as the market price of H₂ reaches approximately 10 USD per kg H₂.

As shown in Fig. 6(b), the functionality of CO₂ in line with the distribution of carbon is well-supported by the three phase products mass balances. The GPR induced by CO₂ with volatiles evolved from SMS led

to carbon reallocation from pyrogenic oil to pyrogenic gas. Taken together, the results of this study prove the mechanistic role of CO₂ in reducing volatiles via GPR. Thus, increased production of CO by GPR offers a strategic measure for enhancing carbon utilization in thermochemical processes.

4. Conclusions

This study emphasizes on the functional use of carbon dioxide for the valorization of SMS into syngas. CO₂ imparts a greener/sustainable feature to the pyrolysis process and simultaneously serves a reactive raw material. The use of CO₂ altered the carbon distribution towards gaseous products. This is attributed to the gas-phase reaction of CO₂ with volatiles (evolved from SMS) enhancing CO formation. To promote the reaction kinetics of CO₂, employing Ni catalyst led to a noticeably increased syngas yield (mainly CO). Thus, the use of CO₂ in the pyrolysis system facilitates the enhanced carbon utilization.

CRedit authorship contribution statement

Taewoo Lee: Investigation, Formal analysis, Writing – original draft. **Dongho Choi:** Investigation, Formal analysis, Data curation. **Jonghyun Park:** Investigation, Formal analysis, Data curation. **Yiu Fai Tsang:** Formal analysis, Data curation. **Kun-Yi Andrew Lin:** Resources, Formal analysis, Data curation. **Sungyup Jung:** Data curation, Writing – review & editing. **Eilhann E. Kwon:** Supervision, Conceptualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

This work was supported by a National Research Foundation of Korea (NRF) grant funded by the Korean government (MSIT) (NRF-2023R1A2C3003011).

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